

Md Shahedur Rahman

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Professional Summary:

Collaborative, self-driven, and detail-oriented professional with extensive experience in the creation and optimization of strategic IT solutions. Versatile across various aspects of software development with a strong emphasis on teamwork and innovation. Committed to staying up to date with the latest trends and technologies to fuel continuous professional growth.

Key Skills:

- Designing software architecture, developing software applications, conducting comprehensive testing, and meticulously debugging code to swiftly identify and rectify any anomalies or challenges
- Technical Skills: Python | Django | React | SQL | Tableau | Power BI | Azure Databricks | Kafka | Excel | NumPy | Pandas | Apache Airflow | Jupyter Notebook | JavaScript | Java | C++ | Vue.js | Spring Boot | JDBC | Java ORM | Rest Services | Boot Strap | Angular | Thymeleaf | Database Design | Table Design | PostgreSQL | E/R diagrams | Agile | Unit Testing (JUnit) | Integration Testing | Git | IntelliJ | Flask | SciPy | ETL Development | Web Scraping | Data Visualization | Data Analytics
- iOS Development Skills: Swift, Xcode, UX/UI Design, RESTful API, and Apple SDKs
- Cloud Computing Skills: AWS, Azure, IBM, Google Cloud Service and Cloud Deployment Models (IaaS, PaaS), Kubernetes, Docker, Cloud CI/CD Pipeline, and Microservice Development

Projects:

- eCommerce website as a collection of RESTful services. **(Python/Django/Docker/Flask)**
Designed and developed backend applications using Python with Django and Flask, implementing RESTful services and enhancing the Wishlists module with optimized query and action routes. Ensured code quality and reliability through comprehensive Python unit tests in Django. Built and tested REST APIs with Django REST Framework, integrated Swagger for API documentation, and containerized applications using Docker. Managed automated CI/CD pipelines with GitHub Actions and Jenkins, deploying services to a Kubernetes cluster on IBM Cloud, while adhering to Agile methodologies with sprint planning and continuous integration.
Project Link: <https://github.com/NYU-DevOps-Summer-2022/wishlists>
- CloudIdentify (Multi-Cloud Project). **(Python/Microsoft Azure/AWS)**
Designed and developed cloud-native applications with a strong focus on Microsoft Azure and Python to build a multi-cloud learning platform for real-time image processing, data storage, and analysis. Leveraged Azure Computer Vision API, Azure Blob Storage, Azure Functions, AWS Lambda and GCP to implement scalable, serverless workflows and integrated them with Python Flask for the backend. Built and optimized pipelines for secure data ingestion and analysis, incorporating best practices in object-oriented design, SQL-based relational databases, and CI/CD pipelines using GitHub Actions and Docker for containerization and deployment.
Project Link: <https://github.com/MDSShahedurRahman/MultiCloudProject>
- MetaML, a peer-to-peer, decentralized platform. **(Python/Django/Flask/asyncIO)**
Collaborated with a team of three software developers to develop a peer-to-peer, decentralized platform with a focus on Open World Applications such as competition platforms. Contributed to the Knowledge Management functionality for managing reusable software components. Utilized Python and various libraries including Kademlia for distributed hash tables, Django for backend development, Flask for REST framework implementation, asyncIO for concurrent and asynchronous code, Scipy for optimization and signal processing, and MySQL for database management.

- Twitter Sentiment Analysis for U.S. and Australian Elections. (Python/SQL/Power BI/Kafka)
In this project, I developed a robust data engineering pipeline to process and analyze tweets from the social media platform Twitter, focusing on public sentiment towards political parties in the U.S. 2020 Presidential Election and the Australian 2019 Election. Leveraging Python and SQL, I built scalable ETL workflows to ingest and transform large volumes of tweet data. Using Apache Kafka, I implemented real-time streaming pipelines for ingesting live Twitter feeds, and stored and processed the data in Azure Databricks for distributed analytics. I employed Excel for initial data exploration and validation, while Tableau and Power BI were used to design interactive dashboards that visualized sentiment trends and their potential impact on election outcomes.
- VeloCity - A City Tours Optimization App. (Java/Spring Boot/React/Vue.js/PostgreSQL)
VeloCity is an innovative web application that streamlines city tours by integrating advanced travel planning technologies. It enables users to explore city landmarks effectively, offering features like real-time landmark availability, advanced searching options, optimized travel routes, and personalized itineraries. The platform incorporates Google APIs such as Distance Matrix and Places API for accurate route and landmark data, leveraging a Traveling Salesman Problem (TSP) algorithm to optimize routes for minimal travel time. Built using Java, Spring Boot, Vue.js, Angular/React, JavaScript and PostgreSQL.
Project Link: <https://github.com/MDShahedurRahman/Capstone-veloCity>

Certificates/Training/Education:

- **NEW YORK UNIVERSITY (NYU)** | New York, United States Jan '21 – Dec '24
Master of Science in Computer Science
- **TECH ELEVATOR (Remote)** | Cleveland, OH, United States May '24 – Aug '24
Java Web Application Developer Bootcamp
- **CODE PATH (Remote)** | San Francisco, California, United States Feb '24 – May '24
iOS Developer Bootcamp
- **PER SCHOLAS** | Bronx, New York, United States Apr '23 – Jul '23
Full Stack Java Developer Bootcamp
- **UNIVERSITY OF CALIFORNIA (UCLA)** | Los Angeles, California Sep '12 – May '15
Bachelor of Science in Physics

Professional History:

Altice USA Inc / News12 | Edison, New Jersey, United States Dec '19 – Apr '22
IT Engineer

- Developed and maintained IT automation solutions using Python and Shell scripts to streamline and accelerate daily operational responsibilities including automated User Account creation
- Ensured the smooth execution of television broadcasting tasks and inhouse network operations by promptly diagnosing and resolving 99.99% of equipment issues within an average of 10 minutes
- Designed, installed, and evaluated a comprehensive range of broadcasting and IT equipment, encompassing 10+ file servers, robotic cameras, satellites, and audio-video consoles

Regional News Network (RNN) | Rye Brook, New York, United States Sep '15 – Nov '19
IT Engineer

- Created, tested, and maintained Python and Shell scripts for routine system administration activities including analyzing current IT operations and identifying opportunities for automation
- Configured and maintained specialized equipment for video production, broadcast, satellite transmission, and interactive media, ensuring uninterrupted operations across multiple locations, resulting in a 30% reduction in connection issues
- Minimized service disruption by quickly identifying and implementing alternative solutions when equipment failed